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LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY

LOURDES HILLS, KUTTICHAL P. O., THIRUVANANTHAPURAMM - 695 574

ANSWER BOOK

Name Asiya Shalul Hameed Class No 12 Branch MCA

Subject Data Science and Machine Learning Semester S3

Mark Tabulation

Mark Divisions	Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	Q.14	Q.15
	3	3	3	3	3	3	6	6							
Total															
Grand Total													30		

1. Data Science

* It is an ~~extendi~~ interdisciplinary field - I focus on extracting knowledge from dat Sets. which are typically huge in amount

* The field analysis and presenting findi to inform high level decisions in an organiza

* Such incorporate skills from Computer Scie mathematics, statistics information visualizati graphic and busines

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Excellent

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* data is everywhere and is one of the most important feature of every organization that helps a business to flourish.

* Data Science and data analysis helps us to ensure we get answers for the questions from data

* It is the Concept to bring together ideas data examination Machine learning and their related strategies

need of data Science

* we have more human data and more numerous tools for gathering warehousing and interpreting it but the need for data Science is growing frequently and perceived as essential in many diverse Sectors

* It is one of the most popular emerging domain and most sought after career option : Analyst

* data Science can add value to any business who can use their data well for statistics and insight

12. Machine Learning :

* Machine learning is the field of study that gives Computers the capability to learn without being explicitly programmed.

* ML is one of the most exciting technologies that one would have come across.

* It makes Computers more similar to humans ability to learn

* It is an essential skill for any aspiring data analyst and data Scientist and also for those who wish to transform a massive amount of raw data

* ML is a type of artificial intelligence that allow the Software applications to become more accurate at predicting outcomes without being explicitly programmed to do

* It is for building Mathematical model and making predictions using historical data or information.

* It is using for vaious tasks like image recognition Speech recognition. email filtering facebook. auto-tagging etc.



Classification method

clustering

Regression Models

dimensionality Reduction

Deep learning etc

Very good Answer!

data with labels

I/P

I/P data without label

I/P static and ac

Supervised learning

Unsupervised learning

Reinforcement learning

error

Critic

output Mapping

output classes

output (state/actions)

3 Euclidean formula

$$D(P,q) = \sqrt{(P_1 - q_1)^2 + (P_2 - q_2)^2}$$

where P and q are feature of given class. This can be used to classify data by finding matching neighbors

tomato (Sweet = 6, Crunchiness = 4)

Here, $P_1 = 6$
$q_1 = 4$

grape (Sweet = 8, Crunchiness = 5)

$P_2 = 8$
$q_2 = 5$

$$D(P,q) = \sqrt{(6-8)^2 + (4-5)^2}$$

$$= \sqrt{(-2)^2 + (-1)^2}$$

$$= \sqrt{5}$$

∴ This is distance between tomato and grape

4. Conditional probability Bayes theorem

* naive Bayes algorithm is based on Bayes theorem Bayes theorem gives the condition probability of an event given another event B has occured

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

P(A|B) = Conditional probability of A given
P(B/A) = Conditional probability of B given
P(A) = Probability of event A
P(B) = Probability event B

eg: tossing a coin

{HH, HT, TH, TT}

H - Head ; T - tail

P(Second Coin is head given first coin is tail) = P(A|B)

= [P(B/A) * P(A)] / P(B)

= [P(first Coin is tail given Second Coin is head * P(Second coin is head) / P(First time tail)

= (1/2) * (1/2) / (1/2)

= (1/2)

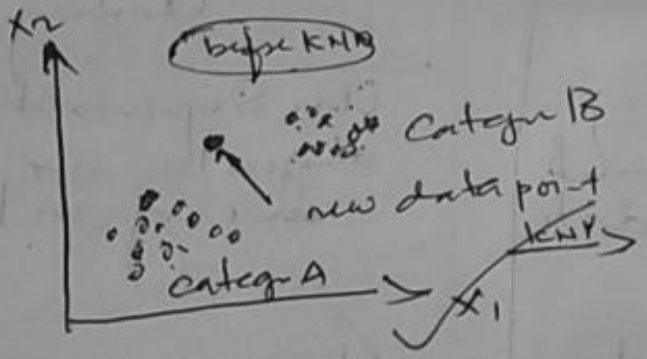
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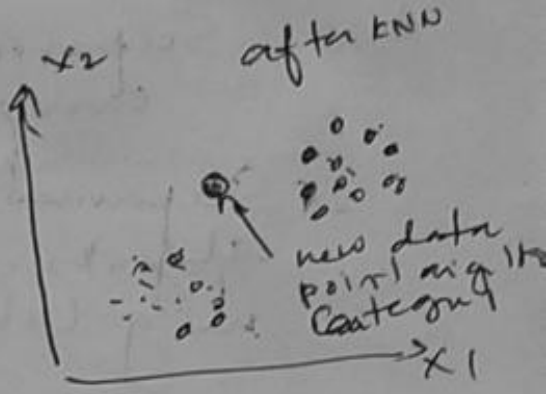
ADDITIONAL SHEET

5. KNN

* K-Nearest Neighbor is one of the Simplest Machine learning algorithm based on Supervised learning technique.

* K-NN algorithm assume the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories





Work of KNN

Step1:- Select the number of K neighbor

Step2:- Calculate the euclidean distance of K number of neighbor

Step3 : Take the K nearest ... the calculated ... algorithm

Step4 :- among the K neighbours the number of the data point in each category

Step 5 : Assign the new data point to that Category for which the number of the neighbours is maximum

Step 6 : Our model is ready.

6.

Supervised learning	Unsupervised learning
* Supervised learning algorithms are trained using labeled data	* Unsupervised learning algorithms are trained using unlabeled data
* If a model output takes direct feedback to check if it is predicting correct output or not	* If is a model does not take any feedback

ADDITIONAL SHEET

* @ model is predict the output	model finds the hidden pattern to data
* it is a input data is provided to the model along with output	* it only input data is provided to the model
* It's goal is to train the model so that it can predict the output when it give new data	* its goal to learn to find the hidden pattern and useful insight from the unknown dataset
* It categoried in classification and Regression	* it is clustering and association

Eg:

ADDITIONAL SHEET

7.

$P(yes) = 5/10$

$P(no) = 5/10$

orgine	yes	no
domstic	3/6	3/6
Imported	2/4	2/4

Color	yes	no
Red	3/5	2/5
yellow	2/5	3/5

Type	Yes	no
SUV	1/4	3/4
Sport	4/6	2/6

$P(yes) \times P(Red/yes) \times P(domstic/yes) \times P(SUV/yes)$

$= \frac{5}{10} \times \frac{3}{5} \times \frac{3}{6} \times \frac{1}{4}$

$= 0.5 \times 0.6 \times 0.5 \times 0.25$

$= 0.0375$

$P(no) \times P(Red/no) \times P(domstic/no) \times P(SUV/no)$

$= 5/10 \times \frac{3}{5} \times \frac{3}{6} \times \frac{3}{4}$

$= 0.5 \times 0.4 \times 0.5 \times 0.75$

$= 0.075$

$P_{NB}(Vel) = \frac{0.875}{0.875 + 0.375}$

$= 0.333$

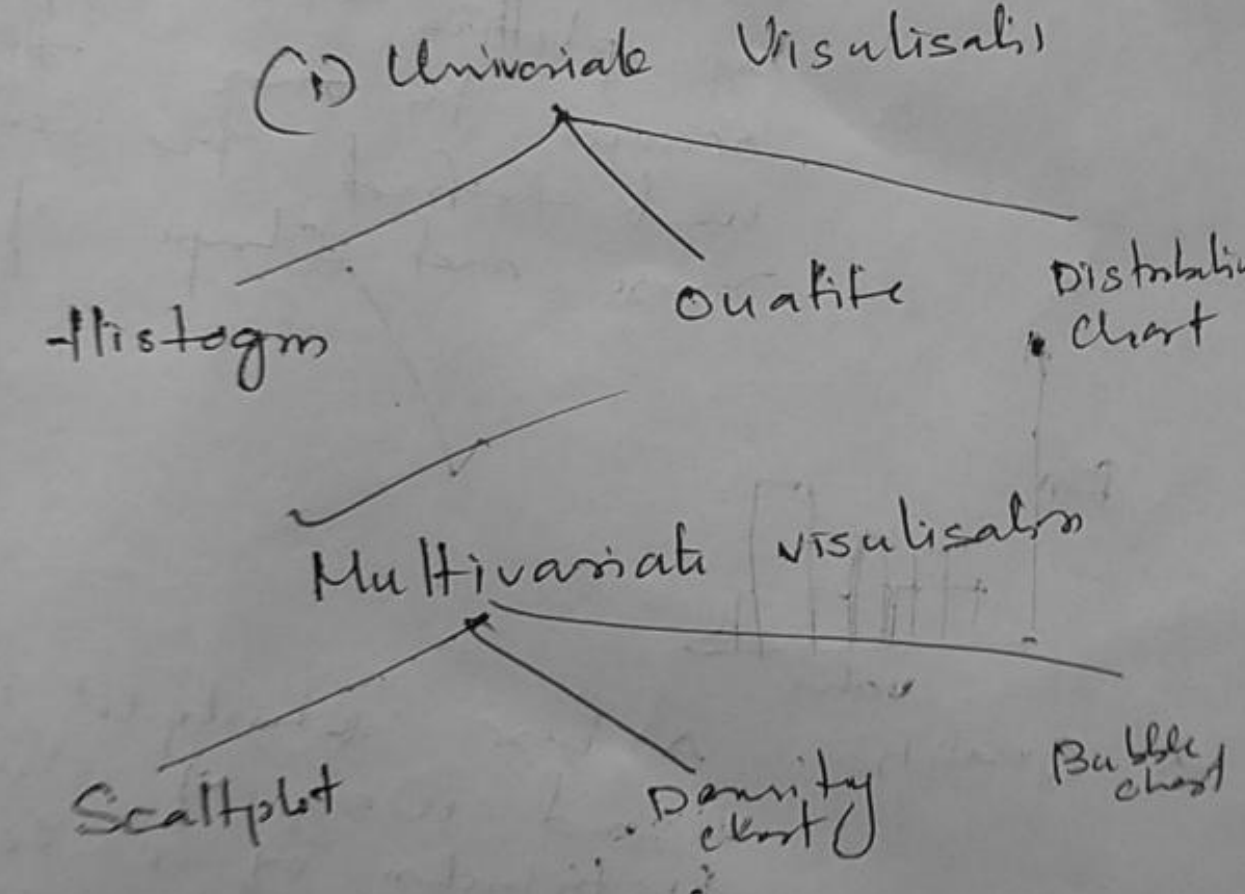
$P_{NB}(No) = \frac{0.875}{0.875 + 0.4375}$

$= 0.666$

The above probility we can
that the can with given
not stolen

9. Data Visualization is the one of the most important technique

- it discovery of data data discovery and explorations this visual representation of data provide easy comprehension of complex data with multiple of attributes



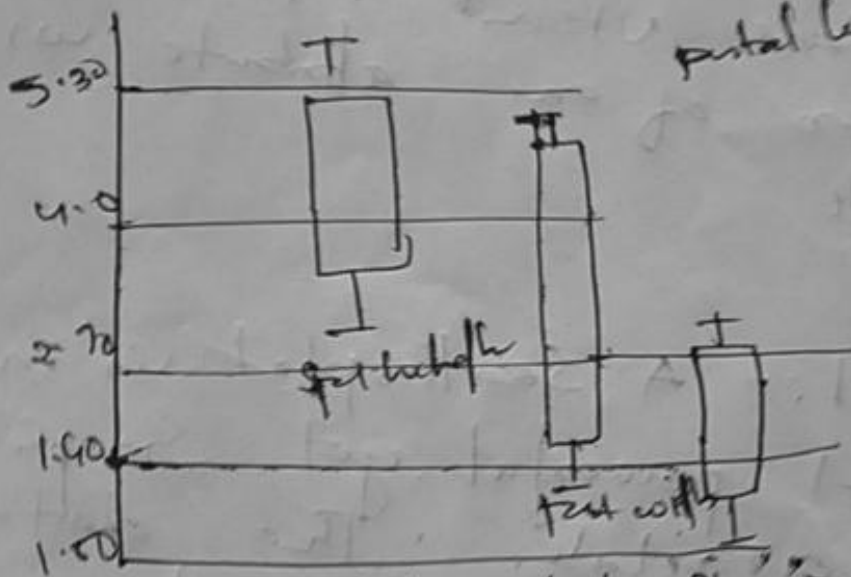
Univariate Visualisation

* Visual exploration involving the investigation of one variable at a time using univariate visualisation techniques

* Histogram : A histogram is one of the most basic visualisation techniques to understand the frequency of the occurrence of values. It shows the distribution of the data by plotting the frequency of occurrences in each histogram bin. We can find the central tendency and shape of distribution



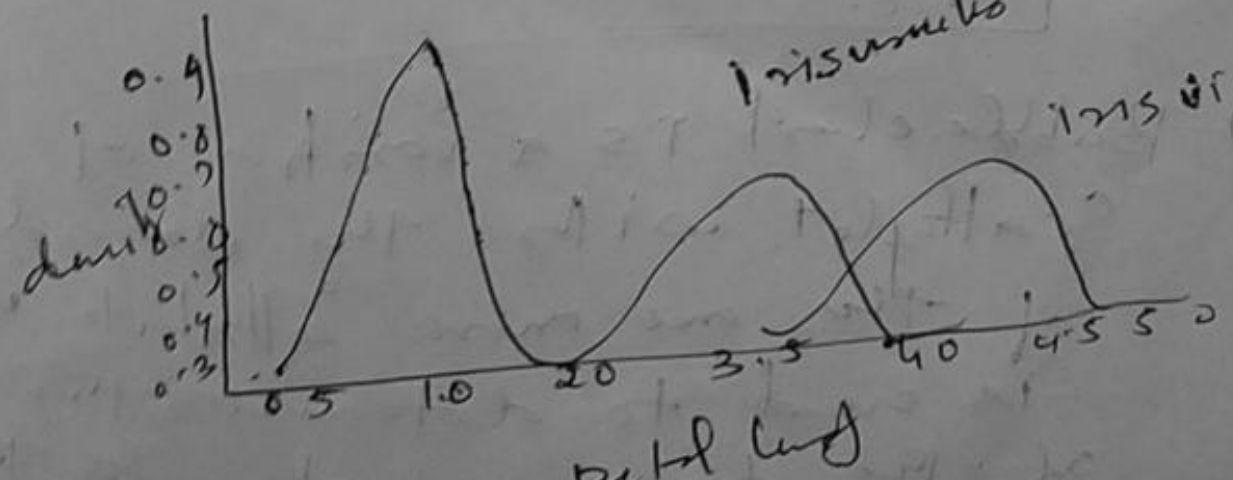
* Quartile : A box whisker plot is a single visual way of showing the distribution of a continuous variable with its quartile measures such as quartile median and outliers. Quartiles are measured by dividing the



Quick plot of iris dataset

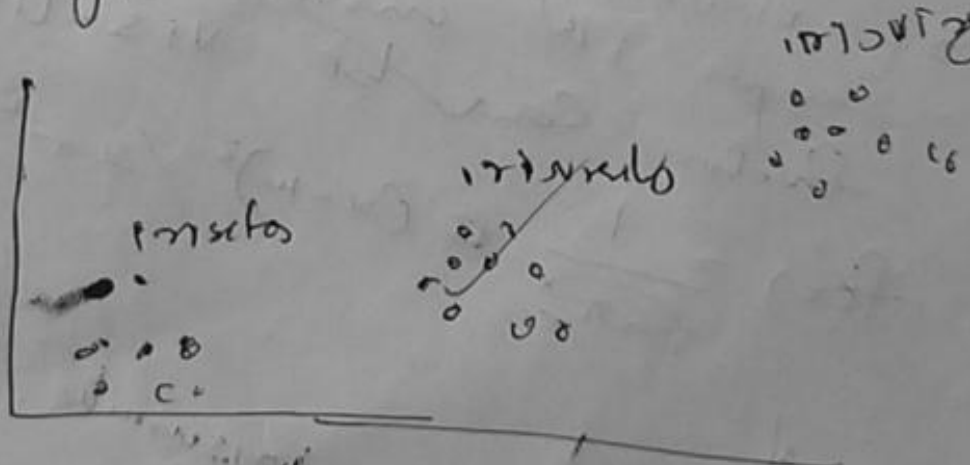
C. distribution chart: for Continuous variables like petal length in the dataset visualizing the actual data in the Sample it's normal distribution function can be visualized in the

$$f(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{(x-\mu)^2/2\sigma^2}$$



Multivariate Visualization : visual explo
Considers more than one attribute in
the same visual . It focus on the
relation of one attribute with anot
attribute

a) Scatterplot :- A Scatterplot is one of th
most powerful yet simple visual
plot available . The data point
are marked in Cartesian Space
with the attributes of the data
set are only of Continuous data
type



b) "Bubble chart" is a variation of Simple
Scatterplot with the addition
of the one more attribute which
is used to determine the size
of the data point . In the Iris
data set petal length and petal
width are used for x and y axe



e) Density Chart : are Similar to Scatterplots with one more dimension as background color. The data points also be colored to visualize one more dimension and a total of four dimensions can be visualized in a density



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